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DEVELOPMENT OF A RETRIEVAL-AUGMENTED GENERATION BASED RESEARCH MONITORING SYSTEM FOR INTELLIGENT THESIS TRACKING SYSTEM]

A Capstone Project Proposal
Presented to the Faculty of the
Polytechnic University of the Philippines
Unisan, Quezon Campus

In Partial Fulfilment of the Requirements for the
Diploma in Information Technology

by

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CERTIFICATION OF ORIGINALITY

This capstone project "***DEVELOPMENT OF A RETRIEVAL-AUGMENTED GENERATION BASED RESEARCH MONITORING SYSTEM FOR INTELLIGENT THESIS TRACKING SYSTEM***", prepared and submitted by **LEONARD C. ADIQUE, SAYRAGIN B. AYAG, CHRISTINE JOY C. BERNABE, JENNILYN M. CUETO, JOVIN N. JUEZAN, DEXELL O. JUSI, JOHN MICHAEL L. NADAL, & FLOR SHAIRA M. ORIEL**, in partial fulfillment of the requirements for the, DIPLOMA IN INFORMATION TECHNOLOGY has been examined and recommended for Oral Examination.

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CERTIFICATION OF ORIGINALITY

This is to certify that the research work presented in this capstone project, *DEVELOPMENT OF A RETRIEVAL- AUGMENTED GENERATION BASED RESEARCH MONITORING SYSTEM FOR INTELLIGENT THESIS TRACKING SYSTEM* for the Diploma in Information Technology at the Polytechnic University of the Philippines embodies the result of original and scholarly work carried out by the undersigned. This capstone project does not contain words or ideas taken from published sources or written works that have been accepted as basis for the award of a degree from any other higher education institution, except where proper referencing and acknowledgment were made.

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ABSTRACT

Title	:	Development of a Retrieval-Augmented Generation based Research Monitoring System for Intelligent Thesis Tracking System
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(Note: The Abstract is a brief description summary of the thesis, capstone or dissertation. It contains the main objective of the study, a brief description of the research method, major findings, conclusions or main arguments, and recommendations. Abstract should be doubled – spaced, and not more than 250 words.)

This study developed a Research Monitoring System with built-in data analytics to improve how research activities are managed and tracked at PUP Unisan, Quezon Campus. The system addresses common issues in the manual process, such as slow retrieval of research records and limited monitoring. Data analytics tools, including summaries and dashboards, were added to support clearer and faster decision-making.

A mixed-method approach was used. Interviews helped identify user needs, while a survey based on selected ISO 25010 criteria evaluated the system's functionality, usability, and reliability. The results show that the system works as intended, is easy to use, and provides useful insights. Overall, it offers an efficient and adaptable way to strengthen research management.

Keywords: Polytechnic University of the Philippines, Diploma in Information Technology, Research Monitoring, Data Analytics, ISO 25010, Decision-Making, System Evaluation

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Research is fundamental to society, shaping areas like healthcare, education, technology, and social policy. It involves a structured process of inquiry aimed at addressing problems, generating knowledge, and deepening our understanding of the world. Research is crucial for advancing society and fostering the creation of new ideas, technologies, and innovations (Francie Atienza, 2023). It continues to drive advancement across various disciplines by shaping perspectives, addressing complex challenges, and fostering innovation (Mendonca, 2009). Over the decades, research has significantly contributed to societal growth by producing breakthroughs that make life more convenient as more advanced topics emerge.

In line with technological advancements, Research Monitoring Systems have become essential in ensuring the smooth operation of digital infrastructures and research-related applications. These systems aid organizations in detecting issues, optimizing performance, and maintaining system integrity. In fog computing environments, monitoring plays a crucial role in orchestrating services by gathering real-time status information from fog nodes and communication links. This allows the orchestrator to make informed decisions to maintain service-level agreements, such as redistributing workloads to more capable nodes or optimizing service placement based on historical performance and failure patterns.

Similarly, data analytics has become a major trend across institutions. It involves examining raw data using various methods to extract meaningful insights, with many of its techniques now automated through algorithms and computational processes. Data analytics strengthens organizational performance and supports decision-makers by

providing evidence-based insights (The Investopedia Team, 2025). Its ability to collect, organize, interpret, and present information greatly improves outcomes and contributes to positive impacts across society.

Retrieval Augmented Generation (RAG) improves an AI model's performance by linking it to external knowledge sources. Through this approach, large language models (LLMs) can generate more accurate, relevant, and higher-quality responses. A typical RAG pipeline works in three steps: Extraction, where data is collected, transformed, indexed, and stored in a vector database. Retrieval is where the data is identified and retrieves relevant data using vector and keyword search techniques. Generation, is where the Large Language Models (LLMs) combine the user's prompt with retrieved data to craft answers that are simultaneously semantic and contextually precise (NVIDIA, n. d.)

Monitoring research within Higher Education Institutions (HEIs) presents several challenges, especially in campuses that rely on manual or semi-digital processes. Many universities struggle to track the progress of student and faculty research effectively, as key aspects such as data quality, monitoring procedures, realignment of objectives, and documentation often become difficult to manage consistently (Silva et al, 2024). These issues affect the ability of research coordinators and advisers to ensure compliance, maintain accurate records, and oversee the integrity of research activities. Outdated and fragmented monitoring processes further intensify these problems.

Studies show that the use of inefficient digital or manual systems can lead to delays in retrieving records, data inaccuracies, and difficulty tracking academic progress—issues similarly observed in other institutions like Philippine Merchant Marine Academy (PMMA) (Reyes & Hernandez, 2021). Such limitations create frustration among stakeholders, with students experiencing delays in receiving feedback and

educators struggling to monitor progress efficiently (Garcia, 2020). These concerns reflect the broader challenges faced by many state universities and colleges.

These issues are also evident in the context of the Polytechnic University of the Philippines Unisan, Quezon Campus, where research monitoring is still conducted through traditional means. The absence of centralized and real-time monitoring system results in slow communication between students and advisers, risks of misplaced documents, and inconsistencies in tracking revisions or compliance. Without an automated and unified system, the research advisers and coordinators encounter challenges that hinder efficient supervision and reporting. These concerns highlight the need for a dedicated Research Monitoring System that can streamline workflows, enhance traceability, and support evidence-based decision making within the campus.

This study aims to design, develop, implement, and evaluate the software quality of a smart Research Monitoring System with Data Analytics and Retrieval-Augmented Generation that will streamline and support the monitoring tasks of researchers in PUP Unisan, Quezon Campus. The goal is to help users track their compliance with research activities more efficiently. The researchers will develop a web-based system intended to assist research coordinators and advisers, while also helping students stay on track, produce quality work, and develop effective research habits. Additionally, the study seeks to evaluate the system in terms of functional suitability, interaction capability, and reliability. The system is expected to improve adviser-student communication, especially when problems or inconsistencies arise. It will also ensure that research projects progress properly, reducing incomplete tasks and preventing inconsistencies. Furthermore, the system will support adherence to academic and ethical standards by ensuring confidentiality, proper consent and development of clear achievable research objectives.

This research contributes to broader institutional goals related to educational equity, data privacy, and work ethic. By developing a monitoring system integrated with data analytics, the study aims to help reduce gaps in research monitoring and support more equitable access to academic resources. The project also contributes to several Sustainable Development Goals (SDGs), particularly Quality Education, by promoting inclusivity, equitable learning opportunities, and lifelong learning. It supports Industry, Innovation, and Infrastructure by encouraging creativity and technological advancement; Peace, Justice, and Strong Institutions through transparency and accountability in research monitoring; Partnerships for the Goals by fostering collaboration; Decent Work and Economic Growth by promoting efficient and ethical research practices.

Theoretical Framework

This study is grounded in the Information System Success Theory developed by Delone and McLean (1992). This theory states that the success of an information system depends on several key factors: the quality of the information provided, the quality of its services, how often and how effectively it is used, the satisfaction of the users, and the overall benefits it brings to an organization. Together, these key factors offer a framework for evaluating the performance of a system in terms of usability and impact. In this study, the model serves as the primary basis for assessing the Research Monitoring System imbedded with Data Analytics intended for PUP Unisan, Quezon Campus.

The theory's major concept-system quality, information quality, service quality, use, user satisfaction, and net benefits describe how the proposed system should function. System quality refers to the system's performance, reliability, and user-friendliness. Information quality involves accuracy, relevance, and timeliness of research data and analytics produced by the system. Service quality pertains to the stability and support users experience whilst interacting with the platform. The concept of Use focuses on

how students, advisors, coordinators, and administrators utilize the features such as progress tracking, updating status, adding remarks, and viewing research records. User satisfaction reflects their overall experience and improvements users perceived compared to the traditional monitoring process. These factors influence net benefits, which includes enhanced communication, faster monitoring, improved documentation, and more organized research information. These concepts work together and influence one another. When system, information, and service quality are high, users are more likely to use the system consistently and be satisfied with it. This increase usage and satisfaction lead to stronger net benefits for both individuals and institution. This pattern deeply reflects the desired outcomes of the Research Monitoring System, where reliable performance and accurate data handling are expected to improve usability and make research monitoring more efficient.

This theory aligns with the study's objectives which involves designing, developing, implementing, and evaluating a system that monitors research activities and provides a web-based platform for tracking progress, sharing feedback, updating statuses, assigning research categories, and generating structured information. The IS Success Model supports this aim by offering criteria and determining whether the system is reliable, produces quality information, promotes effective use, satiety users, and brings meaningful improvements such as better documentation and reporting. Its alignment with ISO 25010 is clear, as both frameworks focus on evaluating system performance through aspects such as functionality, reliability, usability, and information quality. While the IS Success Model highlights user satisfaction and overall benefits. Together, they help make sure the system is assessed in terms of both user experience and technical quality.

This theory is highly relevant to the study because it provides a coherent foundation for evaluating academic information systems, especially those used for

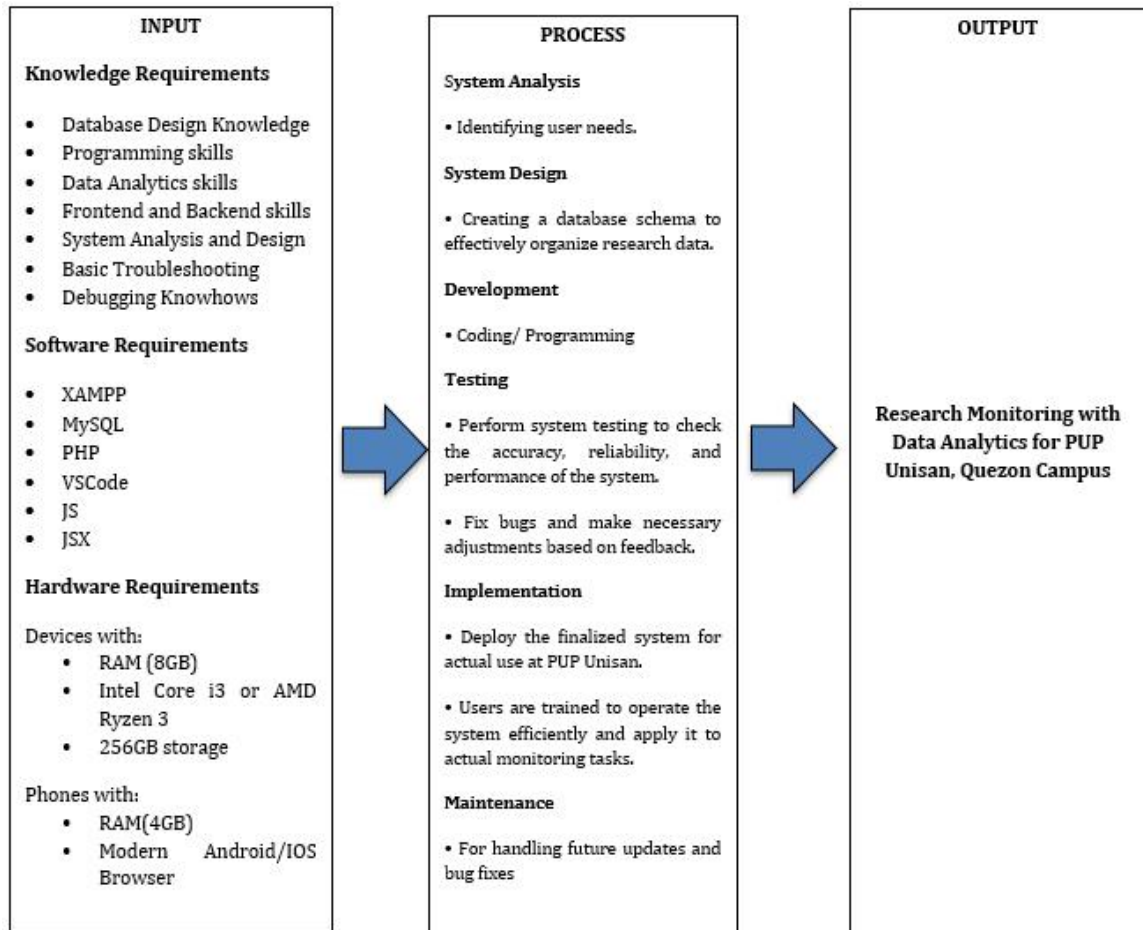
research monitoring. It ensures that the Research Monitoring System with Data Analytics is assessed not only on its technical functions but also on its ability to deliver accurate information, enhance communication, improve workflows, and support institutional research processes. Through this theoretical lens, the system is positioned as a practical solution to the challenges currently faced in research monitoring at PUP Unisan, Quezon Campus

Conceptual Framework

The conceptual framework illustrates the flow and relationship among the key components involved in developing a Research Monitoring System with Data Analytics for PUP Unisan, Quezon Campus. It follows the Input–Process–Output (IPO) model.

The development of the Research Monitoring System with Data Analytics for PUP Unisan, Quezon Campus, begins with the collection of essential inputs that serve as the foundation of the system.

Figure 1: Input Process Output (IPO) Chart



The inputs required for developing the Research Monitoring System with Data Analytics include various technical, knowledge-based, software, and hardware components. The project requires essential knowledge and skills in database design, programming, frontend and backend development system analysis and design, data analytics and basic troubleshooting. These competencies enable the developers to design the system, build its features, and fix issues during development. The project also needs different software tools, such as XAMPP for server setup, MySQL for managing the database, PHP and JavaScript for building the system, JSX for interface components, and VSCode as the main coding application. For hardware developers need a laptop or desktop with the minimum of 8GB RAM, an Intel i3/ AMD Ryzen 3 processor, and 256GB of storage to properly run and test the system. Meanwhile, users will need mobile

devices with at least 4GB RAM and updated Android or IOS browsers to access the system smoothly.

During the process phase, the system undergoes several stages of development. It begins with system analysis and design, where user needs are identified and a database schema is created to organize research data effectively. Next comes the system development phase, where the actual coding and implementation of modules for data entry, tracking, and reporting take place. Once developed, data integration and analytics tools are incorporated to allow the system to collect and analyze research information, providing meaningful insights and trends. The system then proceeds to testing and evaluation to ensure that it functions correctly, is user-friendly, and meets the desired objectives. Finally, deployment and training are conducted so users can operate the system efficiently and apply it to actual research monitoring tasks. Additionally, maintenance will be implemented when new features are added and errors inevitably occur, to keep the system flow smoothly and advanced.

The output of this process is a fully functional Research Monitoring System with integrated data analytics. This system allows PUP Unisan to manage research activities more efficiently, improve transparency and accessibility of data, and provide valuable insights that support academic and institutional decision-making. It ultimately enhances the quality and productivity of research within the university.

Statement of the Problem

The research monitoring process in Polytechnic University of the Philippines – Unisan, Quezon Campus is mostly done manually, which makes it difficult to track

research progress, organize data, and generate reports. This often results in delays, errors, and lack of updated information for administrators, faculty, and students.

In the context of Data Analytics in Monitoring Research, the manual monitoring of advisors of research under their guidance is tedious and time consuming, considering that students must go to their advisor's vicinity and advisors are often, not, inundated by their workload. However, the primitive way of manual monitoring causes unnecessary situations, such as having unaligned schedules and advisors having to check other student's research work. This is problematic because inconsistent monitoring can cause myriads of errors overlooked, and this can affect the paper grimly. While previous research has discovered ways in monitoring research work that could create lasting impacts in a respective field such as, keeping sure that objectives are clear, systematic data collection, engaging stakeholders and others. But the gap is that it lacks contextual studies, academic factors thus not offering much help for the current and future researchers.

General Objectives

To design, develop, implement, and evaluate a system that monitors research works, and to provide a web-based platform that allows students and advisors to track research progress, exchange feedback, update statuses and remarks, assign research categories and access accurate and organized research records. The system will also enable users to generate structured research information that supports efficient documentation and reporting.

Specific Objectives

1. To analyze existing research monitoring systems and identify their limitations.

2. To design and implement a proposed research monitoring system tailored to PUP Unisan, Quezon Campus.
3. To integrate data analytics and retrieval-augmented generation (RAG) to enable intelligent querying and generate insights on research trends, contributions to SDGs, and performance metrics.
4. To evaluate the system's performance using criteria such as usability, functionality, and reliability.

Research Questions

Specifically, the project aims to answer the following questions:

1. What limitations or gaps exist in the existing research monitoring systems?
2. How can a research monitoring system be designed to meet the specific needs of PUP Unisan, Quezon Campus?
3. How can data analytics be incorporated into the research monitoring system to track research trends and performance metrics?
4. What is the effectiveness of the proposed system in terms of usability, functionality, and reliability?

Scope and Limitations of the Study

This study aims to design, develop, implement, and evaluate a research monitoring system embedded with data analytics to track students' progress in their research work and an AI feature to know everything that a student needs to know from how to apply for a specific document to applying to copyright. The system focuses on research outputs of students enrolled in the Diploma in Information Technology (DIT) and Bachelor of Science in Information Technology (BSIT) programs at the Polytechnic University of the Philippines, Unisan, Quezon Campus, starting from Academic Year

2024. The system will focus solely on monitoring the statuses of research outputs—specifically Title Approved, Proposal Approved, UREC Forms Submitted, UREC Clearance Released, Final Defense Completed, and Copyright/ Intellectual Property Submitted, excluding the actual submission of full manuscripts through the system. The system will not cover research outputs of faculty members; instead, it is limited to student research projects only. Furthermore, the system will track research progress solely within the PUP Unisan campus, excluding students or research collaborations outside the campus. In addition, the data analytics features will focus on generating insights on research trends, contributions to SDGs, and performance metrics, while it will not perform advanced predictive analytics or AI-based research assessment.

Significance of the Study

This study will address the absence of an efficient and centralized monitoring system for research activities at PUP Unisan, Quezon Campus. Further this study will provide a modern solution through the development of a Research Monitoring System integrated with Data Analytics that will enable real-time tracking, organized data storage, and insightful analysis of trends and productivity in research.

The following groups are identified as the primary beneficiaries:

Higher Educational Institutions. The study benefits Higher Educational Institutions by centralizing research information, improving monitoring efficiency, reducing administrative workload, and supporting accreditation efforts. With data-driven insights and better documentation, institutions can enhance transparency, collaboration, performance, and decision-making.

PUP Unisan, Quezon Campus. PUP Unisan, Quezon Campus will directly benefit since the system is designed for its research processes. The platform enables easier monitoring and organizing of student research outputs, while the data analytics provides

insights into productivity, trends, and areas needing support. This strengthens documentation, boosts efficiency, and supports the campus in maintaining academic excellence.

Research Coordinator. Research coordinators benefit from simplified tracking of proposals, progress, and completion status. The system reduces manual paperwork, centralizes research information, and speeds up communication between coordinators, advisers, and researchers.

Research Advisers. Advisers can monitor their advisee's work, revisions, and deadlines more effectively. With consolidated information and analytics, advisers can quickly identify students who need assistance and observe common manuscript issues, making communication and feedback faster and consistent.

Student Researchers. Student researchers gain a clear and organized platform for tracking progress, revisions, deadlines, and feedback from their adviser. Data analytics help them understand common mistakes, and areas for improvement. Faster communication and clearer requirements help reduce confusion and support timely completion of research work.

Researchers. Current researchers benefit from a more organized workflow with unified submissions, deadlines, and adviser feedback. The addition of analytics provides insights into trends, common errors, and areas needing improvement. Streamlined monitoring helps reduce delays, confusion, and stress.

Future Researchers This study contributes to future research by demonstrating how data analytics can enhance institutional research management. It serves as a reference for campuses or organizations aiming to integrate analytics into their monitoring systems, supporting improved evaluation process and greater transparency.

By promoting evidence-based governance, capacity-building, and innovation within higher education, the study supports several UN Sustainable Development Goals,

including SGD 4 which is Quality Education through improved academic system and accountability; Industry, Innovation and Infrastructure by encouraging the use of advanced technologies and research innovation, Peace, Justice and Strong Institutions which fosters transparency and institutional effectiveness. And Partnerships for the Goals that enable better data sharing, research coordination, and data sharing

Definition of Terms

This section provides clear explanations of key terms and concepts used throughout the study.

Academic Research — a systematic, critical investigation by scholars and students to expand knowledge, solve problems, or understand phenomena that involves collecting and analyzing data.

Analytic Tool — refer to software application and methods that enable organizations to collect, process, and analyze data to gain insights and make informed decisions.

Data Analytics — is a multidisciplinary field that involves the collection, transformation, and organization of data to draw conclusions, make predictions, and drive informed decision-making.

Research Monitoring — refers to the process of overseeing research activities to ensure they are conducted in accordance with approved protocols, regulatory requirements, and ethical standards.

Retrieval-Augmented Generation (RAG) — an AI technique where an external data source is connected to a large language model (LLM) to generate domain-specific responses in real time.

Chapter 2

REVIEW OF LITERATURE AND STUDIES

This chapter delves deeply into what is already proven about research monitoring. It focuses on the existing literature and studies that have a direct correlation to the development of Research Monitoring System with Data Analytics. It examines the current trends, issues, and theoretical foundation relative to research monitoring and information systems. The chapter identifies gaps of previous works and establishes the basis for designing a system suited to the needs of PUP Unisan, Quezon Campus.

Review of Existing Research Monitoring System and Their Limitations

This section of the study delves into the existing research monitoring systems and their possible gaps. Researchers will determine the existing systems' advantages and the recurring gaps that are worth bridging. This segment is about identifying the issues and understanding why the study is being conducted.

Research Monitoring System involves evaluating commercial and open-source platforms that help academic institutions, research libraries, and showcase research activities. System monitoring is an important basis for system modelling and improvement. Monitoring and evaluation are essential components of the research process, as they help maintain the quality, integrity, and timely completion of research outputs. Establishing an effective monitoring system is crucial for strengthening research governance (Enciso, 2024).

Research Monitoring Systems such as Research Information Management Systems (RIMS) are essential for supporting not only recording and monitoring research activities but also the implementation of structured and in-depth evaluation processes. By serving as a centralized platform for storing research metadata, these systems assist

in decision-making related to resource distribution, performance evaluation, and institutional planning. Additionally, they help improve transparency, collaboration, and overall efficiency within the research environment (Schöpfel & Azeroual, 2024). Another monitoring system is the Current Research Information System (CRIS), which typically includes data on researchers, projects, outputs and funding resources. To fully utilize the extensive research metadata stored in CRIS systems across different institutions and countries, standardized methods for data exchange are required. These interoperability mechanisms enable the aggregation of information, providing a broader and more comprehensive view of research activity within a particular field beyond a single institution (de Castro, 2020)

Nevertheless, these monitoring systems have limitations that are an opening to conduct such study. These limitations often make researchers have problems in monitoring surveillance. Watson et al. (1997) noted that Information Systems departments have long faced complex challenges due to rapidly changing environments and increasing organizational demands. These challenges often require IS managers to identify and address key issues within system design and implementation. Similarly, Disadvantages of Management Information System (n.d.) highlights that MIS solutions themselves are inherently complex and costly, requiring clear communication between organizations and software developers to ensure that system requirements are properly understood. Together, these sources emphasize that both the organizational environment and the systems being developed contribute to ongoing difficulties in IS management.

Concluding this section, it is evident that existing research monitoring systems possess capabilities that assist users and researchers in managing data and supporting evidence-based decision making. However, these systems still present issues and gaps that make monitoring processes challenging. Recognizing these limitations, the

researchers identified an opportunity to address the gap, creating a new system incorporating data analytics to enhance the system's effectiveness.

Principles and Approaches in Designing Research Management System

In this segment of the study, the focus is on the principles and approaches in designing the Research Monitoring System tailored for the Polytechnic University of the Philippines (PUP) Unisan, Quezon Campus. The researchers will determine which design approaches are appropriate to apply and what principles should guide the system's development. This section emphasizes the importance of establishing a well-structured and effective research monitoring system design.

The created Research Monitoring System was crafted as a prototype for Higher Education Institutions (HEIs) in the Philippines to improve the administration and supervision of research outputs. The system features a data input module (Fig. 1A) for recording details from research papers, such, as application type, paper title, publication type and date venue, abstract, keywords and the research document. An additional module (Fig. 1B) handles information about authors and co-authors. Links the research environment to the relevant organization matching the institution's research goals. These components guarantee data handling, aid in swift data access, for evaluation and reporting and assist in tracking research operations. The system prioritizes user-friendliness, adaptability and aiding decision processes enhancing research oversight (Regla and Ballera 2024).

A procedure was created to assist in making choices about managing research data points to maintain data accuracy while reducing mistakes. Every warning produced by the system is. Given a code according to its correctness. When a data point is identified as incorrect it is labeled "00." For entries the system contrasts the value against past patterns, in the research database. If later data show that the value lies

within permitted limits it is labeled “01” This method allows for tracking of research results helping administrators detect irregularities, ensure precision and facilitate well-informed decisions. Kovach et al, (2011)

The Model–View–Controller (MVC) is a design pattern, in software engineering frequently utilized to create user interfaces by dividing an application into three linked elements. The model handles the data. Contains methods to fetch add and modify data in the database. The view serves as the user interface usually shown as a web page allowing users to engage with the system. The controller functions as a mediator facilitating communication between the model and the view while also overseeing the resources required to handle user requests and produce the interface display (Duldulao, 2024).

Planning is the initial stage of the System Development Life Cycle (SDLC). To create a system that is user-friendly and streamlines the monitoring process, we evaluated existing manual operations during the planning phase. In the analysis phase, management policies, practices, and reports were examined to gain a thorough understanding of the outing monitoring activities. By reviewing the current system, we were able to identify its strengths and weaknesses and clearly recognize the issues that need to be addressed [8]. Additionally, the most suitable hardware and software were considered to ensure the effective and reliable development of the E-College Monitoring System.

Monitoring refers to the continuous and systematic observation of a system to identify changes in its operations. It involves regularly collecting and measuring data over time using appropriate tools so that patterns and variations can be accurately compared. The process goes beyond simply watching what happens—when abnormalities or deviations from the expected flow occur, monitoring allows the system

or user to take corrective action to maintain normal and efficient operation (Stangl et al., 2016).

In conclusion, the principles and approaches in developing research monitoring systems provide a structured framework for data collection, storage, and monitoring. While challenges remain in real-time monitoring and reporting, these principles guide the design of a system that enhances research governance and supports effective decision-making.

Enhancing Research Management through Data Analytics and Retrieval-Augmented Generation (RAG)

This section explains how Data Analytics and Retrieval-Augmented Generation (RAG) can be integrated into a research monitoring system. By embedding these tools into the platform, the system can automatically collect, process, and visualize research data. With these features, it can provide real-time insights, track research progress, detect irregularities, and generate reports that help support data-driven decisions.

According to IBM (2024), embedded analytics helps improve system monitoring by offering real-time tracking and predictive insights that can warn users of delays or possible issues. Chen, Chiang and Storey (2012) also emphasize that visualization tools inside a system help users understand complex data more easily, allowing them to make better and quicker decisions. Wamba et al. (2015) found that systems with built-in analytics increase operational efficiency because they continuously monitor data and provide useful, actionable information. Likewise, Khatri and Brown (2023) explain that software equipped with embedded analytics can support dashboards, automated alerts, and predictive reports, ensuring that research activities are monitored accurately and on time.

Retrieval-Augmented Generation, significantly enhance monitoring systems by connecting Large Language Models (LLMs) to authoritative, live, and proprietary data sources. Allowing the AI to respond in a more highly accurate, context-specific, and specific insights. According to AWS (n.d.) RAG enables the large language model (LLM) to provide precise information along with proper source attribution. The results can include citations or references, and users can access the original source documents for further details or clarification. This feature helps build trust and confidence in the generative AI system. Univio (2025) supported this claim by stating that it is a search-augmented generation works by first searching through corporate documents, technical articles, and other relevant sources before generating a response. This approach enriches the query with the most pertinent information, resulting in more accurate, current, and reliable answers while minimizing the risk of AI “hallucinations.”

In summary, integrating Data Analytics and Retrieval-Augmented Generation (RAG) into a research monitoring system enables raw data to be automatically transformed into meaningful and actionable insights. This enhances efficiency, accuracy, and decision-making, while strengthening the system’s ability to monitor research activities and provide administrators and researchers with timely, data-driven information.

Software Evaluation of System Usability, Functionality and Reliability

The modern era of organizational management and research is fundamentally characterized by the strategic deployment of data-driven systems for monitoring and decision support. This section examines how systems are evaluated in terms of their capabilities and accuracy based on ISO 25010 standards, particularly Usability, Functionality and Reliability. Through these criteria, a system’s overall performance and quality are assessed.

This study applies the ISO 25010 quality model, focusing on Functional Suitability, Performance Efficiency, and Reliability as the key evaluation criteria. The ITS

Academic Information System serves as the case study, and the Analytical Hierarchy Process (AHP) method is used to calculate and analyze software quality based on these criteria. The evaluation begins with the preparation of questionnaire items, which are answered by three groups of stakeholders: experts, students, and developers. The AHP method is then applied to analyze their responses, allowing experts to determine the weighted importance of each criterion. These weights are eventually used to compute the overall quality of the software (Sarworsi et al., 2023).

This paper analyzes the perceived usability of learning management systems (LMSs) and examines how usability affects students' learning experience and effectiveness. In the eleven studies reviewed, participants evaluated the usability of two LMSs—eClass and Moodle—using the System Usability Scale (SUS), which consists of 50 items. A total of 769 students participated. The responses were analyzed to determine usability levels, as well as the reliability and validity of the SUS questionnaire. Findings showed that the LMSs received good usability scores, indicating that systematic usability evaluation can provide valuable insights for improving software functionality, performance, and reliability (Orfanou et al., 2015).

Software quality is a critical factor in determining whether a system operates effectively and satisfies its users. The ISO/IEC 25010 standard provides a structured and globally recognized framework for evaluating software based on key attributes such as Functional Suitability, Usability, and Reliability. These characteristics help determine how well the system performs intended tasks, how easy it is to use, and how consistently it functions under normal conditions (Prameswari & Setyawan, 2023).

Evaluating software quality is essential to understanding whether a system meets user expectations. The ISO/IEC 25010 standard is widely used for software evaluation because it identifies the characteristics and sub-characteristics that define overall software quality. In system evaluation, ISO 25010 serves as a guiding framework in

assessing functionality, reliability, usability, security, and efficiency (Januhari et al., 2015).

Assessing software quality plays an important role in determining whether a system can effectively support user needs. The ISO/IEC 25010 standard provides a systematic way to measure software quality across several attributes, making it a reliable foundation for system evaluation (Efendi et al., 2025).

The examined studies collectively emphasize the importance of using ISO 25010 as a standard for assessing software quality. These studies highlight that software quality can satisfy users when the system meets their expectations. Without this criterion, a system might not generate reliable and trustworthy data, thus gravely affecting user satisfaction.

Synthesis of Reviewed Literature and Studies

Research monitoring systems play a crucial role in managing research activities, supporting evidence-based decision-making, and maintaining research quality within academic institutions. Existing systems, such as Research Information Management Systems (RIMS) and Current Research Information Systems (CRIS), provide centralized platforms for storing research metadata, tracking projects, and facilitating resource allocation. These systems enhance transparency, collaboration, and operational efficiency, yet they still present limitations in terms of usability, interoperability, and real-time monitoring. Recognizing these gaps highlights the need for improved systems that can address challenges in research oversight and data management.

To overcome these limitations, the integration of Data Analytics and Retrieval-Augmented Generation (RAG) into research monitoring platforms can significantly enhance their capabilities. Data analytics allows raw research data to be processed,

analyzed, and visualized automatically, providing actionable insights on research trends, performance metrics, and anomalies. RAG further strengthens the system by connecting large language models (LLMs) to authoritative and up-to-date sources, enabling accurate, context-specific responses with source attribution. Together, these tools enhance efficiency, support informed decision-making, and reduce risks associated with incomplete or misleading information, while fostering trust in the system's outputs.

Evaluating such systems requires a structured approach to ensure they meet user expectations and perform reliably. The ISO/IEC 25010 standard provides a globally recognized framework for assessing software quality, focusing on attributes such as functional suitability, usability, and reliability. Studies applying this standard demonstrate that systematic evaluation not only identifies strengths and weaknesses but also ensures that the system effectively supports users' needs. By combining advanced analytics, AI-driven features like RAG, and rigorous software evaluation, research monitoring systems can achieve higher operational efficiency, provide timely insights, and strengthen governance over research activities.

Chapter 3

METHODOLOGY

This chapter represents the methodology used to carry out the study. It involves the research design, research locale, sources of data, research instrument and other procedures followed in developing and evaluating the Research Monitoring System with Data Analytics. This chapter also describes the ethical considerations observed throughout the research process.

Research Design

The Researchers will use the Developmental Research Design approach. Developmental Research Design by its definition is the systematic process of design, development, evaluation, and establishing an empirical basis aimed at the creation of tools or enhanced models (Ramirez, 2023). Considering the nature of the study, this design is appropriate because the study is creating an innovative system that bridges specific gaps in the application of Information Technology.

With iterative procedures such as analysis, design, development, implementation, and evaluation, the design strictly adheres to the developmental process. Each of these stages makes sure that the output is refined, enhanced, and continuously changing based on the needs and feedback of the users. Developmental in nature, this approach requires validation and system assessment, which involves both quantitative and qualitative components.

Using this approach allows the researchers to translate conceptual frameworks into serviceable prototypes. By doing so, it ensures that the system is both sophisticated and practical in addressing real-world needs.

Research Locale

The researchers study setting of choice will be conducted at Polytechnic University of the Philippines (PUP) Unisan, Quezon Campus. The PUP Unisan, Quezon Campus is a branch that offers number of programs that has thesis and capstone under their curriculum, as it is essential. Located at Unisan Quezon, Brgy. Ibabang Kalilayan. This is justified since PUP Unisan, Quezon Campus keenly conducts thesis and capstone projects across its programs, making it a standard setting for the study that is focused on monitoring research activities between students and advisers. The campus also provides approachable respondents and a relevant academic environment where the proposed system can be examined and evaluated effectively.

The data that this research will be obtained from are the 3rd and 4th year college students under the Diploma in Information Technology and Bachelor of Science in Information Technology, enrolled presently during the Academic Year 2025 – 2026. The total population number of 138 students. Additionally, the Research Coordinator and Faculty Research Advisers will be included, as they're the primary users of the system and play a key role in evaluating its monitoring features.

To ensure that each group of respondents was fairly represented and to avoid any form of bias, this study employed the stratified sampling method. A list of enrolled students under the college program Bachelor of Science in Information Technology and Diploma in Information Technology was obtained from the registrar, and thus, respondents were selected using random-number generator.

To determine the required sample size necessary to achieve the desired output, Slovin's formula was applied. Given the total population of 138 respondents and a set of margin of error of 5%, the sample size was computed as follows:

$$N = 138 / ([1 + (138)(0.05)^2])$$

N = 102.6

The computed value was 102.6, which was consequently rounded up to 103 respondents. In addition to the student sample, all research advisers and research coordinator will be added. Therefore, a sample of 116 respondents will be targeted for selection using the simple random sampling technique previously mentioned.

Sources of Data

The primary data will be sourced directly from individuals actively engaged in the research process and will include the principal users of the proposed system students, faculty members, and the Research Coordinator at the PUP Unisan, Quezon Campus. These people are good data providers because they can explain the specific needs of users for the system's design, give firsthand accounts of the current problems with research monitoring, and finally give feedback on how easy and useful the system is after it has been put into use. The researchers will assess the developed research monitoring system's usability, functionality, and overall efficacy using a structured survey questionnaire. Quantitative information on user satisfaction, usability, performance, and overall system evaluation will be gathered through the survey. Instead, interviews will be used to gather data regarding system requirements and current practices. Interviews will be used to collect qualitative data, including the users' expectations for enhancing the research monitoring process, the difficulties they encounter with the current system, and the particular system requirements they require. A direct connection is necessary to capture the human element and guarantee that the technology produced is relevant and user-friendly.

Secondary data will be obtained to provide essential context, foundational knowledge, and operational indicators for the study. The secondary sources used in this study included official university documents, records, reports, and system logs, all of

which the researchers were given access to through the Research Office. These resources contain verified institutional data that supports the analysis of the current research monitoring process. This includes looking over Research Manual/Guidelines to make sure the system is following the right rules and procedures, looking at past research reports to see how much research is being done right now, and searching system logs or databases for quantitative, historical data on how research is submitted and finished. To assess the current state of research monitoring and provide the operational guidelines and historical data needed to build the data analytics module, it is necessary to look over these documents and records.

Research Instrument

In this section of the study, the tools used in garnering data from respondents and other sources will be represented. In order to achieve the study's objectives, these tools are designed to harbor pertinent information in doing so.

This study will use questionnaires as the main tool for collecting data about the Research Monitoring System. The researchers will also conduct a survey as a means of gathering additional information pertaining to the study objectives.

The use of the questionnaire is supposed to gather systematized and measurable responses from the intended respondents. Some of the questions in the questionnaire will employ a 5-point Likert Scale to gauge their perceptions, experiences, and satisfaction with the Research Monitoring System. The two target respondents are students as well as advisers. The researchers will adopt an existing questionnaire from a study authored by Urera Jr. and Balahadia (2019), as the instrument used is appropriate to the context and aligned with the objectives and SDGs of the study, instead of validating the instrument by using experts.

Similarly, the survey will be conducted to gather additional insights that support the overall assessment of the Research Monitoring System. It will follow the same structured format to ensure consistency and reliability of responses.

The data collected through the survey and questionnaire will be subject to high levels of confidentiality. The names of the respondents will not be disclosed, and all the data collected will not be disclosed outside the research.

Data Gathering Procedure

Data gathering procedure is the systematic process of taking and collecting information that is relevant to this study. In order to accomplish the research objectives, the process ensures that the data gathered are both valid and reliable.

The respondents required for this study consist of students at the Polytechnic University of the Philippines, under the college program Diploma of Information Technology (DIT) and Bachelor of Science in Information Technology (BSIT) that are currently taking their research related activities.

The primary instrument used for data collection was a questionnaire adopted by the researchers as it aligns with the objectives and SDGs of the study. Additionally, the researchers will conduct a survey for additional insights that will overall assess the potential advantages and disadvantages of the prospective system.

The study will commence by securing permission directly from the Campus Director of the Polytechnic University of the Philippines Unisan, Quezon Campus, ensuring that all research activities are appropriately authorized. Once approval is given, cooperation with the research coordinator and advisers will be undertaken so that the schedule for the data collection process will be efficient. Heeding the scheduling, the questionnaire will be uniformly distributed to the selected participants, along with an explicit description of the study's objective and directions for completing the instrument.

Upon completion, the researchers will collect the questionnaires and discreetly attest that all responses are complete. Lastly, the collected data will be consolidated and systematically coded in order to prepare it for analysis using statistical methods.

Participants were knowledgeable about the purpose of the study and assured of the concealment of their responses. Informed consent was acquired aforementioned to participate, and data gathered will be held under strict privacy.

Through the systematic and ethical execution of the above procedures, the study ensures the collection of accurate and relevant data to address the research questions effectively.

Statistical Data Analysis

This section represents the statistical methods that will be used to process and analyze data gathered for the study. The statistical analysis guarantees that the findings properly respond to the questions and evaluate the efficiency of the proposed Research Monitoring System with Data Analytics for PUP Unisan, Quezon Campus. The procedures operated in this chapter allows the researchers to understand responses objectively and generate acceptable conclusions based on the survey outcomes.

The study harvested quantitative data from the students and advisers through an adopted questionnaire using a five-point Likert scale, which measured their insights regarding the system's usability, functionality, and reliability. Likert-type responses are reckoned ordinal data, making them apposite for descriptive statistical techniques that condense central tendencies and distributions patterns.

To examine the collected data, the study primarily used descriptive statistical tools. Frequency counts and percentages summarized the demographic profile of the respondents, while the weighted mean measured their level of agreement on the system's usability, functionality, and reliability. Standard deviation was also computed to

reveal the consistency of their responses. These descriptive measures helped present a flawless overview of user perceptions and assess whether the system met adequate quality standards.

Inferential statistics may also be applied, if necessary, to identify distinctions in perceptions between user groups such as students and advisers or to examine relationships between variables such as ease of use and user satisfaction. Nonetheless, because the study focuses on system development and evaluation, descriptive statistics served as the main basis for analysis.

Data processing was conducted in a coordinated manner. Responses were examined for completeness, encoded into spreadsheet software, and assigned numerical values based on the Likert scale. The dataset was reviewed for possible errors before computation. Excel was used to generate the frequencies, percentages, means, and standard deviations required for interpretation. The analyzed results will be presented in Chapter 4 through tables and narrative descriptions that show key patterns and user assessments. These findings will assist in evaluating the system's overall performance based on the study's objectives.

In summary, the statistical procedures used in this study delivered a systematic and objective way to interpret the data, guaranteeing that the results are trustworthy and aligned with the envisioned standards of usability, functionality, and reliability for the developed system.

Ethical Considerations

This study observed strict ethical guidelines to protect the rights, privacy and safety of all participants, particularly the faculty, staff, and administrators of PUP Unisa, Quezon Campus. The researchers ensured transparency by clearly explaining the study's purpose, data collection methods, possible risks, and supposed benefits before

participation. All respondents participated voluntarily and provided informed consent through signed digital and signed forms

To safeguard privacy, no Personally Identifiable Information (PII) was collected. Responses were coded, kept confidential, and securely stored in an encrypted, password protected repository accessible only to the research team. Participants were informed that they could withdraw from the study any time without any penalty or repercussion. The study posed minimal risk, as data was gathered through non-intrusive surveys and interviews, with all questions designed to avoid sensitive content.

Ethical principles were also applied to in the creation of Research Monitoring System. Only dummy or anonymized data were used during system testing, and all research data were handled in compliance with the Data Privacy Act of 2012. Security measures such as encryption and access controls were incorporated into the system to ensure cautious data management. The analytics features were also designed to avoid bias and guarantee accurate, honest results.

All required approvals were obtained, including permission from the University Research Ethics Committee (UREC), consent from the research adviser, ethical oversight from the research coordinator, and authorization from PUP Unisan administration to distribute surveys and conduct interviews. Study documents and consent forms will be firmly stored for the required retention period and permanently removed afterward to prevent unauthorized access.

Throughout the study, the researchers rigorously adhered to institutional policies, national laws, and standard ethical guidelines to make certain responsible and trustworthy research conduct.

Software Requirement Specification

Write your Section intro here...

Functional Requirements.

Put your contents here...

Non-Functional Requirements.

Put your contents here...

Software Development Life Cycle

The Software Development Lifecycle of the Research Monitoring System with Data Analytics describes the structured process followed in developing the system for PUP Unisan, Quezon Campus. It outlines the methodology and phases used by the team to plan, design, build, test, and deploy the system in an organized and systematic manner.

The development of the Research Monitoring System with Data Analytics follows the Iterative Model. This model was chosen because the project required continuous feedback from students, research advisers, and coordinators throughout development. Requirements evolved through stakeholder interviews, and the system was refined through successive cycles of design, implementation, and testing, making it well-suited to the Developmental Research Design adopted in this study.

The SDLC consists of five phases: Requirements Gathering and Planning, System Analysis and Design, System Development and Implementation, Testing and Evaluation, and Deployment and Maintenance. Each phase represents a key step in transforming the project concept into a fully functioning system. Under the Iterative Model, phases are revisited based on feedback until the system meets ISO 25010 quality standards.

In the Requirements Gathering and Planning phase, the team conducted interviews and analyzed existing manual processes to identify system needs. The

System Analysis and Design phase produced the database schema, DFD, use case diagrams, and UI prototypes using the MVC pattern. During System Development and Implementation, the system was built using PHP, JavaScript, and PostgreSQL, with modules for milestone tracking, UREC document management, data analytics, and multi-role access. Testing and Evaluation was conducted with 116 respondents using an ISO 25010-aligned questionnaire measuring Functional Suitability, Interaction Capability, and Reliability.

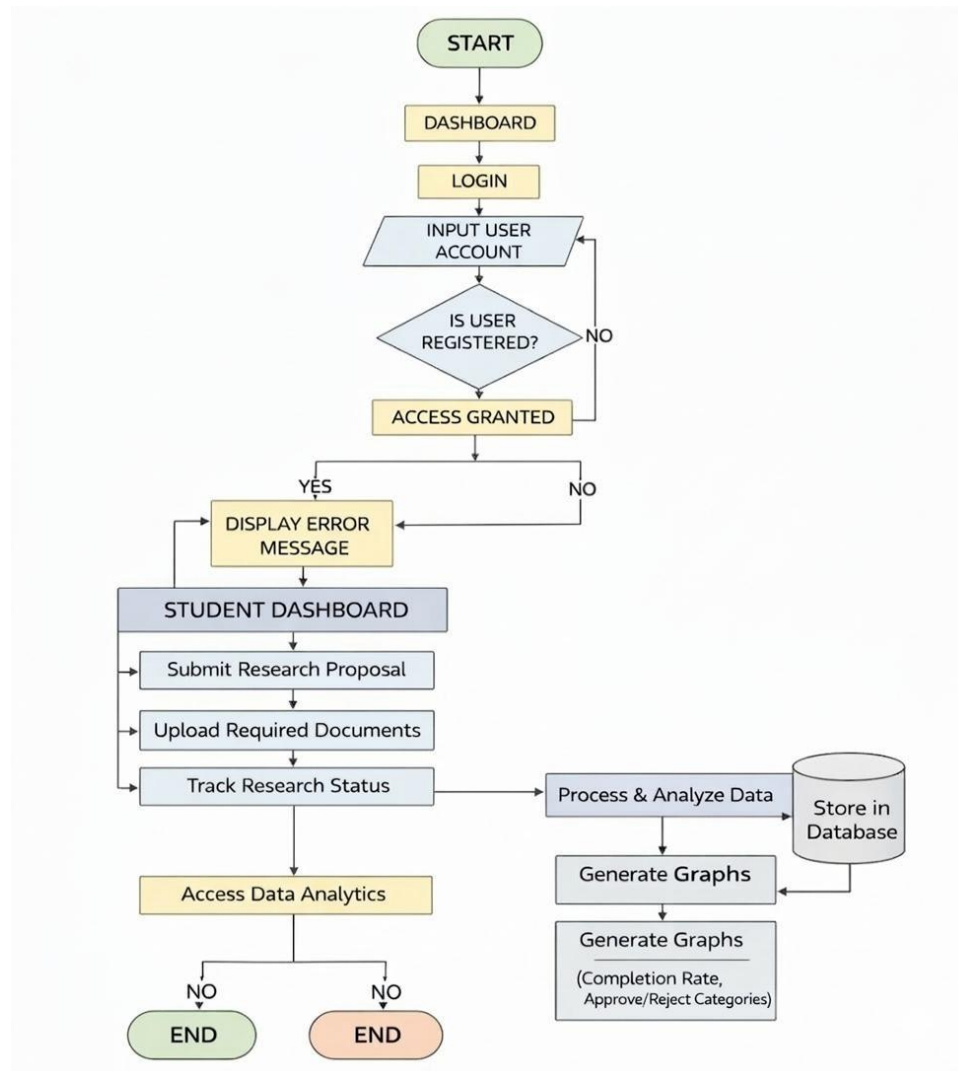
Each phase produces a specific deliverable. Requirements Gathering yields the SRS document and survey instruments. System Design produces the DFD, ER diagram, and UI prototypes. Development delivers the working system codebase. Testing produces the evaluation report and weighted mean tables, while Deployment yields the operational system with backup and maintenance features.

Overall, the Software Development Lifecycle of the Research Monitoring System with Data Analytics provides a systematic framework using the Iterative Model. By following each phase and refining the system based on user feedback, the team ensured the system meets its defined requirements, passes ISO 25010 evaluation, and is ready for deployment at PUP Unisan, Quezon Campus.

System Flow Diagram

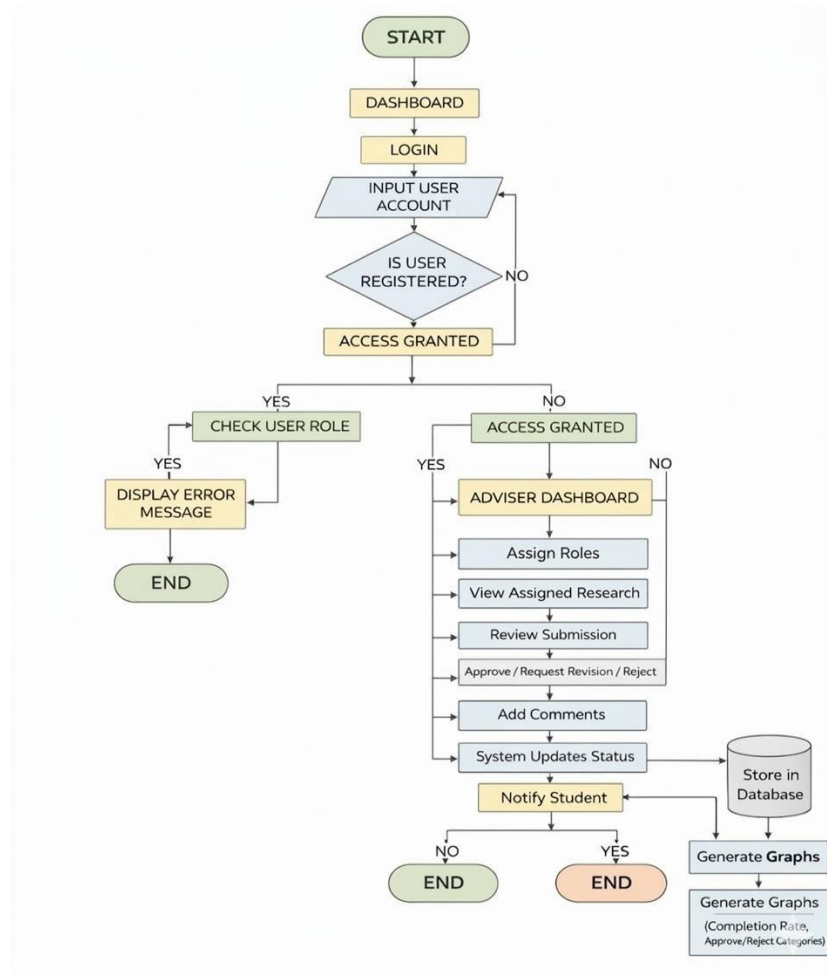
The System Flow Diagram presents a visual representation of how the proposed system operates, showing the sequence of processes and the movement of data within the system. It illustrates the interaction between users, system components, and data storage, providing a clear understanding of how inputs are transformed into outputs. This diagram serves as a guide to the overall workflow of the system, ensuring that each process is logically connected and efficiently organized.

Figure 1: Student System Flow Diagram



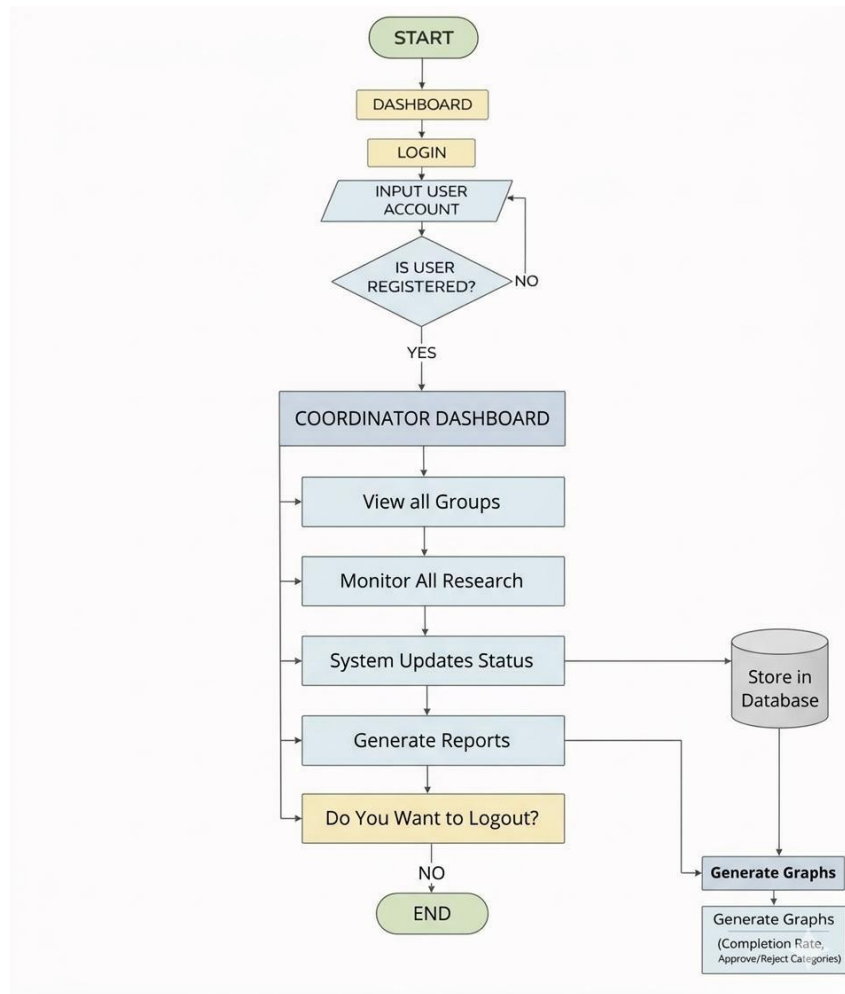
This flowchart illustrates how a student uses the Academic Research Monitoring System. The process starts with logging in, where the system verifies if the user is registered before granting access. Once inside the Student Dashboard, the student can submit a research proposal, upload required documents and track the status of their research. The system then processes and stores the data, generating graphs and analytics to show research.

Figure 2: Advisor System Flow Diagram

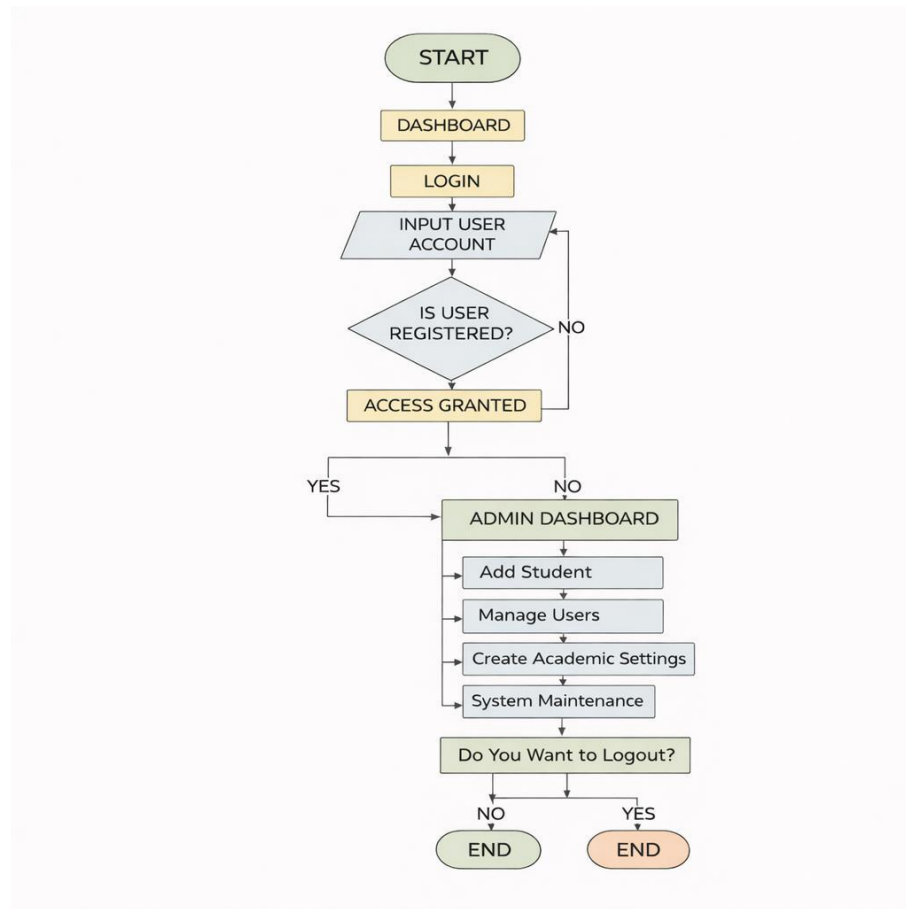


The process ends after the student completes their tasks and exits the system. This flowchart shows how an adviser uses the system. After logging in and being verified, the adviser is directed to the Adviser Dashboard. From there, they can assign roles, review student submissions, approve or request revisions, add comments, and update the research status. All information is saved in the database, which generates graphs to track research progress. The process ends once the adviser finishes and logs out.

Figure 3: Coordinator System Flow Diagram



This flowchart shows how the coordinator interacts with the system. The process begins with logging in, where the system verifies if the user is registered before giving access to the Coordinator Dashboard. Once inside, the coordinator can view all research groups, monitor the progress of all research projects, and update the system status. They can also generate reports based on the collected data.

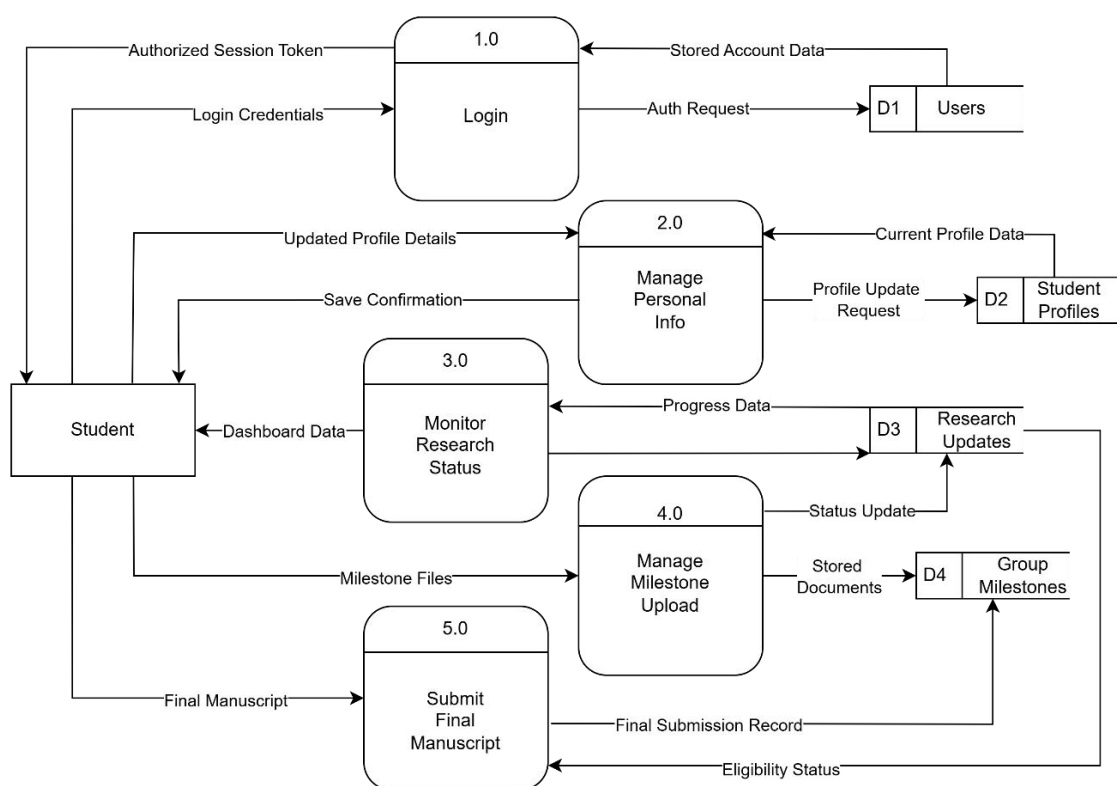
Figure 4: Admin System Flow Diagram

This flowchart shows how the admin uses the system from start to finish. The process begins at the dashboard, where the admin logs in by entering their account details. The system checks if the user is registered, and once verified, access is granted to the Admin Dashboard. Inside the dashboard, the admin can add students, manage users, create academic settings, and perform system maintenance. After completing these tasks, the admin is asked whether they want to log out. Once they choose to log out, the process ends.

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through the Academic Research Monitoring System. It presents the flow of information between users, processes, and data storage, showing how inputs are processed and transformed into meaningful outputs. The DFD helps in understanding the system's structure by identifying the key components and their interactions, ensuring that data is handled efficiently and accurately throughout the system.

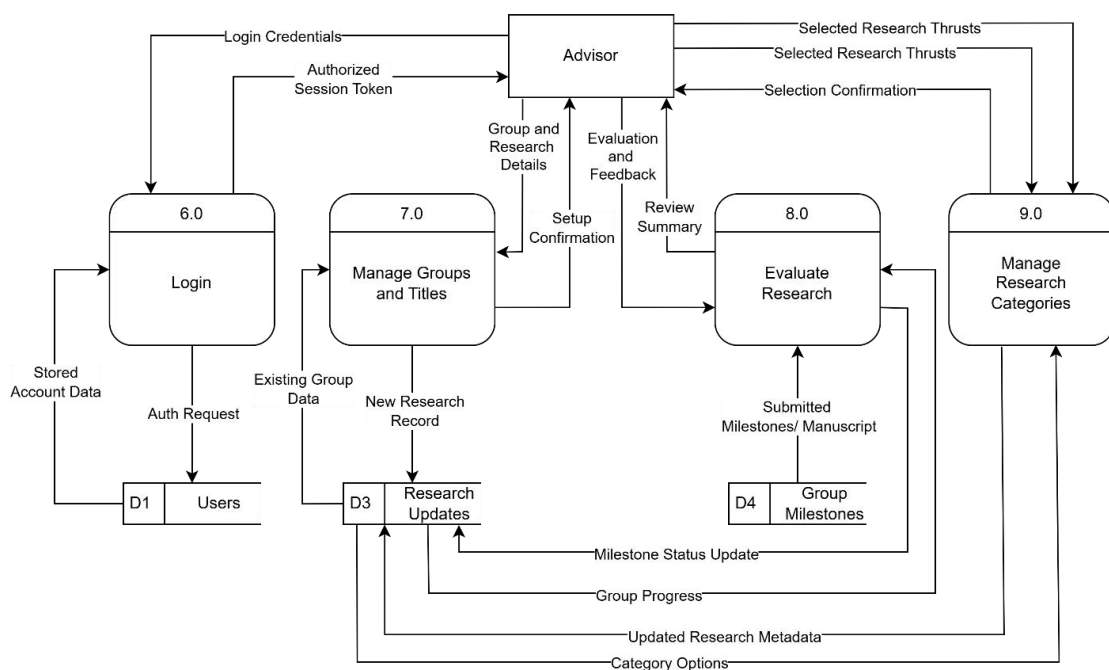
Figure 1: Student Portal



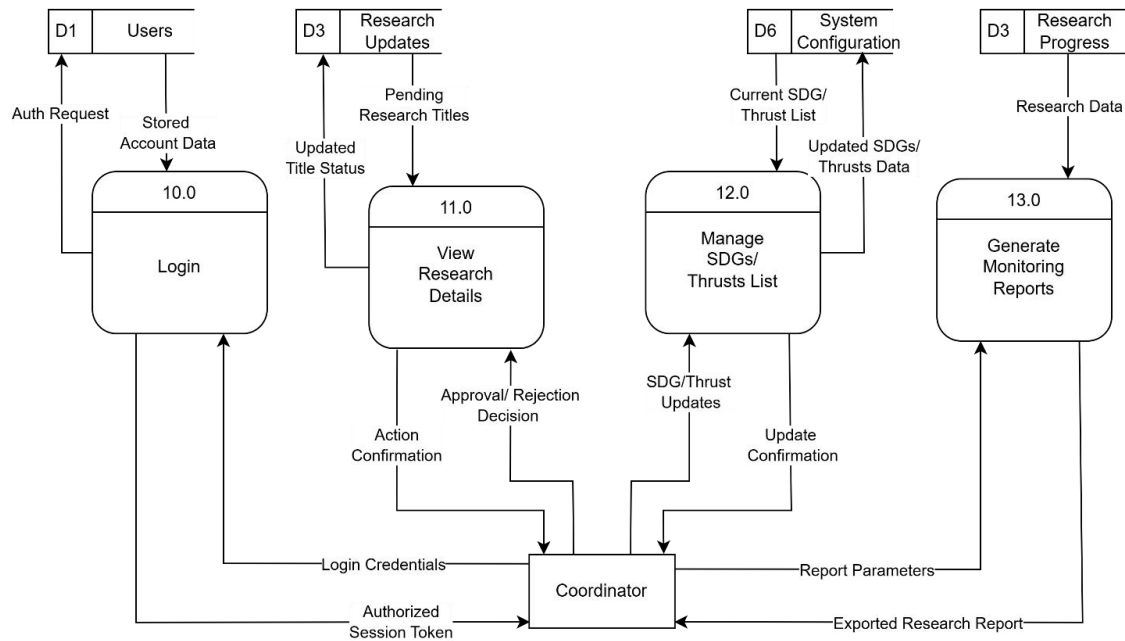
The diagram shows how a student uses the research management system, starting from logging in with their credentials, which the system verifies before granting access. After successfully logging in, the student can update their personal information, and the system saves the changes in the database. The students can also monitor their research progress through the dashboard, where

the system displays updated progress data. In addition, the student is able to upload milestone files, which are stored in the document repository and used to update the research status. Finally, the student submits the final manuscript, and the system records the submission and checks the eligibility status before confirming it.

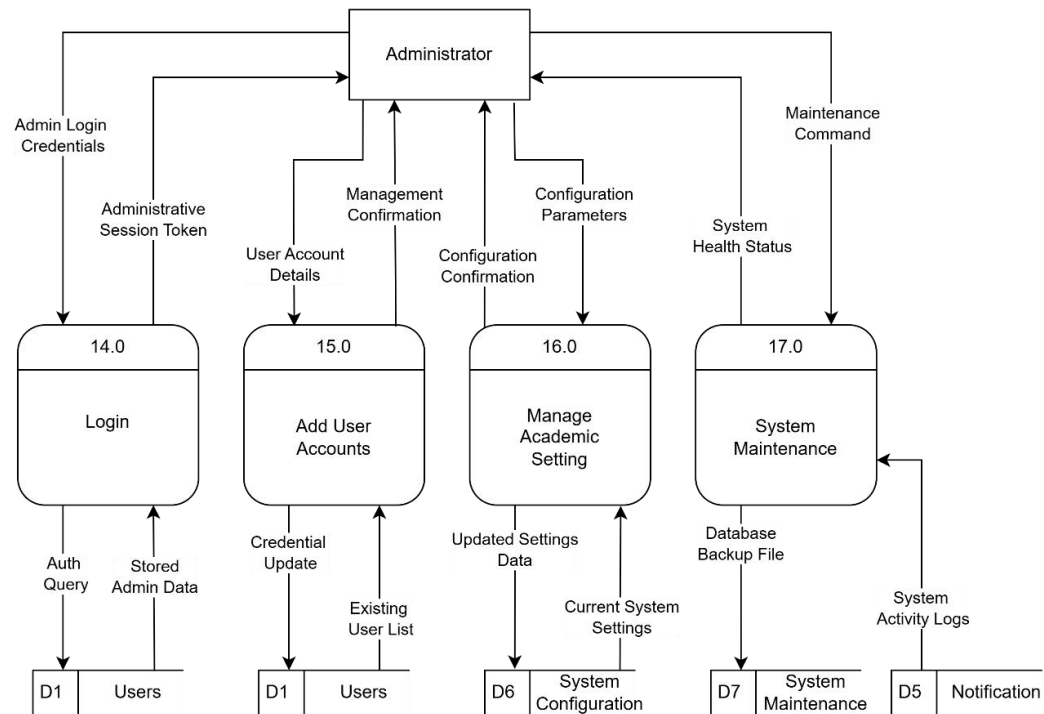
Figure 2: Advisor Portal



The diagram shows how the advisor logs into the system using their credentials, which are verified before access is granted, then manages research groups and titles by viewing existing data and creating new records, evaluates submitted milestones and manuscripts by reviewing documents and giving feedback that updates the research progress, and selects research categories such as UN SDGs and thrusts, with the system saving and confirming all updates.

Figure 3: Coordinator Portal

The diagram shows how the coordinator logs into the system using their credentials, which are verified against stored data before access is granted. After successfully logging in, the coordinator reviews and approves pending research titles, making approval or rejection decisions that update the research progress records. The coordinator also manages system notifications by viewing alerts and clearing or deleting notifications, with all updates saved in the notification log. Additionally, the coordinator generates and exports research reports by selecting report parameters, and the system retrieves research data to produce and deliver the finalized export report, ensuring that all actions and updates are properly recorded in the system.

Figure 4: Admin Portal

The diagram shows how the administrator logs into the system using their credentials, which are verified before an administrative session token is granted. After logging in, the administrator manages user accounts by viewing and updating user details, configures academic settings by modifying and saving system parameters, and performs system maintenance by monitoring system health, creating database backups, and recording activity logs, ensuring that all changes and actions are properly stored and maintained.

Software Prototype

The software prototype of the Academic Research Monitoring System demonstrates the core functionalities of managing and monitoring documents within an organization. It allows users to track document movement, manage records, and

facilitate efficient transfer between departments. This prototype serves as a functional model to evaluate the system's usability and effectiveness in improving document handling processes.

The prototype consists of one main screen: the Login Screen. It allows researchers to access the system by entering their username and password. The Login Screen provides input fields, a login button, a "Remember me" option, and links for password recovery and sign-up. These features help users log in or create an account easily. The user starts from the Login Screen and enters their credentials to proceed. The interface uses simple buttons and fields to guide the user.

Figure 1: Login Screen



Figure 2: Dashboard Screen

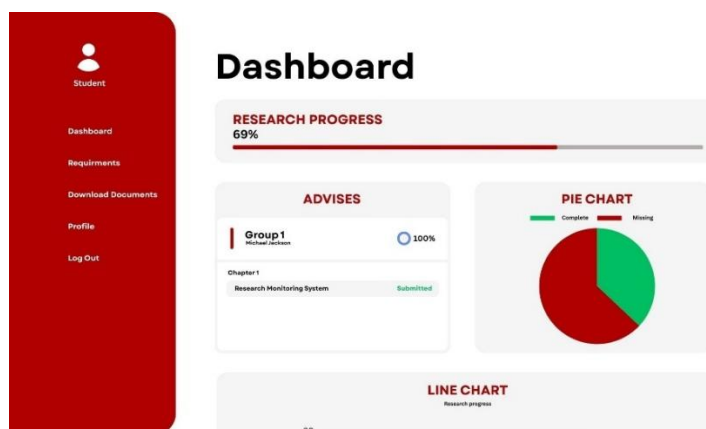
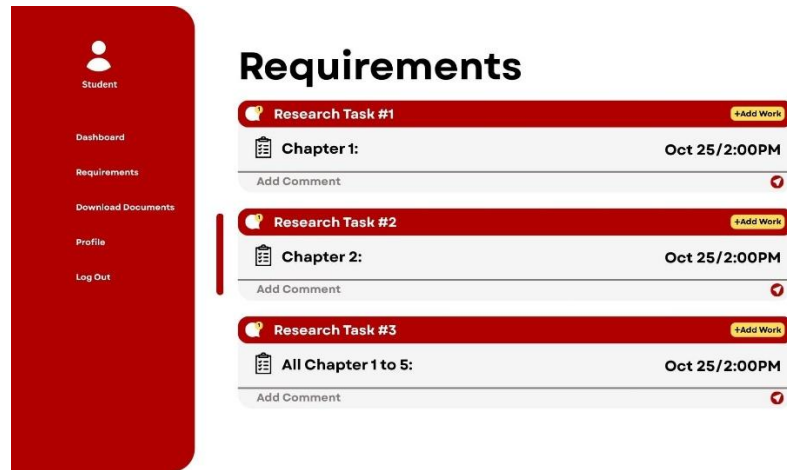
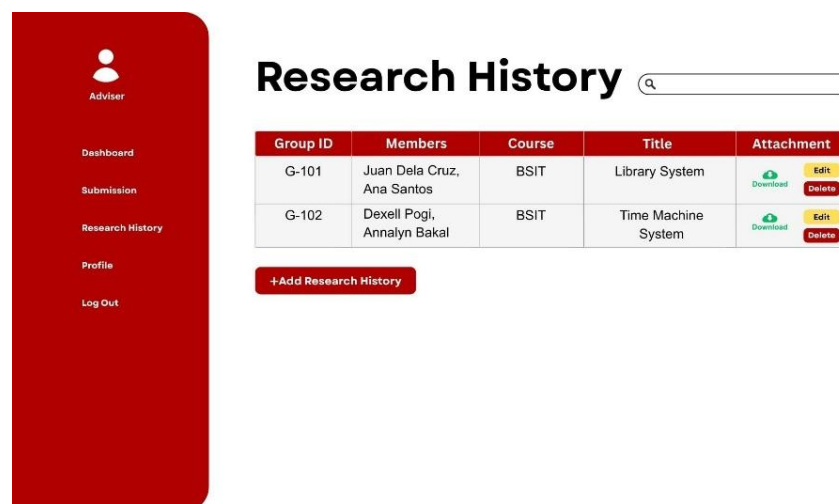


Figure 3: Requirements Screen



The prototype includes the Requirements Screen. This screen allows students to view their assigned research tasks and deadlines. The Requirements Screen displays different research tasks with their corresponding chapters and due dates. It also provides an “Add Work” button and a comment section for submissions and feedback. The user can navigate this screen to check tasks and submit their work. The interface uses simple buttons and labels to guide the user.

Figure 4: Research History



The prototype includes the Research History Screen. This screen allows the adviser to view and manage previously submitted research projects. The Research History Screen displays a table containing group IDs, members, courses, research titles, and attachments. It also provides action buttons such as “Download,” “Edit,” and “Delete” for each record, along with an “Add Research History” button to create new entries. The user can navigate this screen to review past research, update information, or manage files. The interface uses a clean layout with a sidebar menu and simple buttons to guide the user easily.

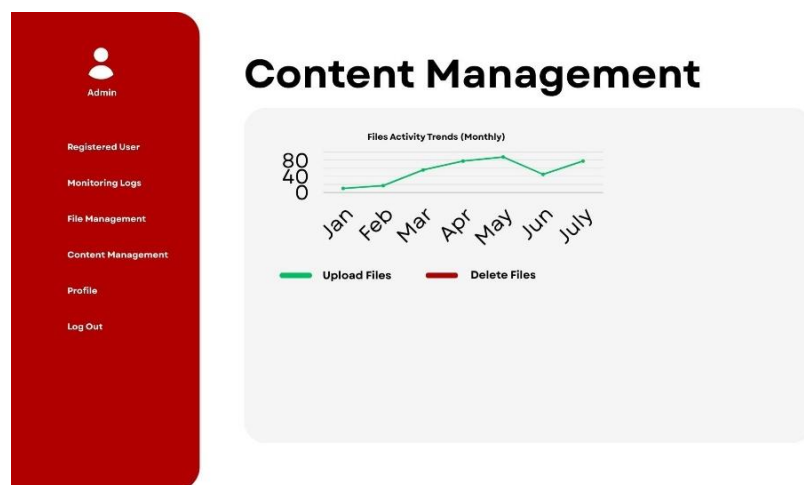


Figure 5: Content Management (Dashboard)

The prototype includes the Content Management Screen. This screen allows the admin to monitor and manage file activities within the system. The Content Management Screen displays a monthly activity graph showing the number of uploaded and deleted files over time. It helps visualize trends in file usage, with labels for each month and indicators for upload and delete actions. The user can navigate this screen to track

system activity and analyze file management trends. The interface uses a simple layout with a sidebar menu and a clear graph to present information effectively.

Project Implementation Plan

The Project Implementation Plan of the Research Monitoring System outlines the structured approach for developing and deploying the system. It presents the phases, tasks, and timeline required to complete the project, serving as a guide for the development team throughout the implementation process.

The implementation plan is divided into three phases: Project Initialization, Prototype Development, and Documentation and Presentation. Each phase represents a distinct stage in the development lifecycle and contains a set of specific tasks that must be completed before progressing to the next phase.

The Project Initialization Phase involves gathering requirements, identifying the problems faced by students and advisers in monitoring research progress, and planning the system prototype features and workflow. During the Prototype Development Phase, the team designs the user interface and system layout, develops the student dashboard, adviser dashboard, and coordinator view, and integrates interaction simulations such as file submission, approval, rejection, and progress tracking. The Documentation and Presentation Phase focus on preparing the project documentation such as the project charter, work breakdown structure, cost outline, and creating the presentation materials for the final system demonstration.

Each phase is assigned a specific timeframe to ensure the project is completed on schedule. The Project Initialization Phase is scheduled during Week 1, the Prototype Development Phase is conducted during Weeks 2 to 4, and the Documentation and

Presentation Phase takes place during Week 5. The overall project is expected to be completed within five weeks.

The project involves eight team members, each assigned specific roles. The Project Manager oversees the schedule, communication, and task delegation. The Lead Designer creates the prototype layout and interface design. The Developers build the prototype pages and simulate system functions. The Documentation Officers prepare the project documents and reports. The Quality Testers check the usability and correctness of the prototype, while the Presenter delivers the final presentation of the system.

Overall, the Project Implementation Plan of the Research Monitoring System provides a clear and organized roadmap for completing the proposed system for the Polytechnic University of the Philippines. It defines the three phases Project Initialization, Prototype Development, and Documentation and Presentation along with their corresponding tasks, timelines, and assigned team members. This ensures that the Project Manager, Lead Designer, Developers, Documentation Officers, Quality Testers, and Presenter can work systematically toward completing and presenting the system on schedule.

Chapter 4

RESULTS AND DISCUSSION

Write an opening section of a chapter that briefly presents the purpose, scope, and organization of the chapter, guiding the reader on what to expect in the succeeding sections.

1. Subheading Number 1

Subheadings in Chapter 4 correspond directly to the findings for each research problem presented in Chapter 1. Each subheading reflects one specific subproblem stated in the Statement of the Problem and is arranged according to the same numerical sequence to ensure consistency and logical flow. These subheadings are formatted in block style, typed in single spacing, and presented in boldface to clearly distinguish them from the body text and to emphasize their importance in organizing the presentation of results.

2. Subheading Number 2

Subheadings in Chapter 4 correspond directly to the findings for each research problem presented in Chapter 1. Each subheading reflects one specific subproblem stated in the Statement of the Problem and is arranged according to the same numerical sequence to ensure consistency and logical flow. These subheadings are formatted in block style, typed in single spacing, and presented in boldface to clearly distinguish them from the body text and to emphasize their importance in organizing the presentation of results.

3. Subheading Number 3

Subheadings in Chapter 4 correspond directly to the findings for each research problem presented in Chapter 1. Each subheading reflects one specific subproblem stated in the Statement of the Problem and is arranged according to the same numerical sequence to ensure consistency and logical flow. These subheadings are formatted in block style, typed in single spacing, and presented in boldface to clearly distinguish them from the body text and to emphasize their importance in organizing the presentation of results.

4. Subheading Number 4

Subheadings in Chapter 4 correspond directly to the findings for each research problem presented in Chapter 1. Each subheading reflects one specific subproblem stated in the Statement of the Problem and is arranged according to the same numerical sequence to ensure consistency and logical flow. These subheadings are formatted in block style, typed in single spacing, and presented in boldface to clearly distinguish them from the body text and to emphasize their importance in organizing the presentation of results.

Chapter 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

Write an opening section of a chapter that briefly presents the purpose, scope, and organization of the chapter, guiding the reader on what to expect in the succeeding sections.

Summary of Findings

Summary of Findings is the section in Chapter 5 of a research study that presents a concise and clear recap of the major results of the study in Chapter 4, based directly on the data analysis, without interpretation, discussion, or recommendations.

Conclusions

Conclusions are the section in Chapter 5 of a research study where the researcher presents general statements or judgments derived from the study's findings, indicating what can be inferred from the results in relation to the research objectives or questions.

Recommendations

Recommendations are the section in Chapter 5 of a research study that present practical suggestions or proposed actions based on the study's conclusions, intended to address identified issues, improve practices, or guide future research..

References

Enter References here

APPENDICES

Appendix 1

PERMIT TO CONDUCT STUDY

Date

Hon. JOHN S. DOE

Municipal Mayor

Municipality of Chicago

Brgy. Dimatagpuan, Chicago, Illinois

Dear Mayor Doe:

We are the 3rd Year students of **Diploma in Information Technology** were tasked to develop a computer system as part of the requirements in **Capstone Project** course. The system aims state here the main purpose of your system.

In line with this, we are requesting your permission to allow us to gather pertinent data with the state here your target group and reason why they are chosen. We assure you that the information gathered will be treated carefully and will be used only for academic purposes.

Thank you for your favorable response and God bless!

Respectfully yours,

MARK JOSEPH C. DELA CRUZ	MARK JOSEPH C. DELA CRUZ
<i>Student Researcher</i>	<i>Student Researcher</i>
MARK JOSEPH C. DELA CRUZ	MARK JOSEPH C. DELA CRUZ
<i>Student Researcher</i>	<i>Student Researcher</i>
MARK JOSEPH C. DELA CRUZ	MARK JOSEPH C. DELA CRUZ
<i>Student Researcher</i>	<i>Student Researcher</i>
MARK JOSEPH C. DELA CRUZ	MARK JOSEPH C. DELA CRUZ
<i>Student Researcher</i>	<i>Student Researcher</i>

Noted by:

NAME OF ADVISER
Capstone Project Adviser

Dr. JONATHAN A. MARQUEZ
Head of Academic Programs

Approved by:

Dr. EDWIN G. MALABUYOC
Director

Appendix 2
INSTRUMENTS

Put your content here....

Appendix 3

CERTIFICATE OF ORIGINALITY

CERTIFICATION OF ORIGINALITY CHECK

This is to certify that the undersigned has been the one responsible for the similarity testing using **TURNITIN** of the capstone project entitled, “**Write your Title Here**” authored by **FirstName M.I. LastName, FirstName M.I. LastName, FirstName M.I. LastName, FirstName M.I. LastName, FirstName M.I. LastName, FirstName M.I. LastName, and FirstName M.I. LastName.**

Furthermore, the combined total of all matches, including overlapping sources, for each database got ____ overall similarity, with a Similarity Report ID: _____.

Certified by:

Head of Admission and Registration

Appendix 4**CERTIFICATE OF EDITING****C E R T I F I C A T I O N**

January 6, 2026

To the **University Research Committee:**

This is to certify that the capstone project entitled **Write your Capstone Title Here** prepared by **Last Name, First Name M.I., Last Name, First Name M.I., Last Name, First Name M.I., Last Name, First Name M.I., Last Name, First Name M.I., Last Name, First Name M.I., and Last Name, First Name M.I.** was examined and proofread by the undersigned person in accordance with the basic mechanics and accepted sentence structures in English.

Dr. PABLO S. GOMEZ, LPT, MEM
Instructor, Polytechnic University of the Philippines
Unisan, Quezon Campus
Language Editor

Appendix 5

ACCEPTANCE LETTER OF ADVISER

Date

Mr. ANGELO P. PERA, MIT

Instructor

Polytechnic University of the Philippines

Unisan, Quezon Campus

Dear **Mr. PERA:**

We are the students under the program of **Diploma in Information Technology** currently taking up our subject, **Capstone Project**. As part of our requirement, we are obliged to have a research adviser. In line with this, we respectfully propose the research topic we intend to pursue for our capstone project: Development of a Retrieval-Augmented Generation based Research Monitoring System for Intelligent Thesis Tracking System.

Given your extensive background in the field of research and information technology, we would like to seek your guidance in crafting our paper and be our research adviser.

Your acceptance and positive response regarding this matter is highly appreciated. Thank you very much and God Bless.

Sincerely yours,

LEONARD C. ADIQUE

Student Researcher

SAYRAGIN B. AYAG

Student Researcher

CHRISTINE JOY C. BERNABE

Student Researcher

JENNILYN M. CUETO

Student Researcher

JOVIN N. JUEZAN

Student Researcher

DEXELL O. JUSI

Student Researcher

JOHN MICHAEL L. NADAL

Student Researcher

FLOR SHAIRA M. ORIEL

Student Researcher

Accepted and Approved:

Mr. ANGELO P. PERA, MIT

Instructor

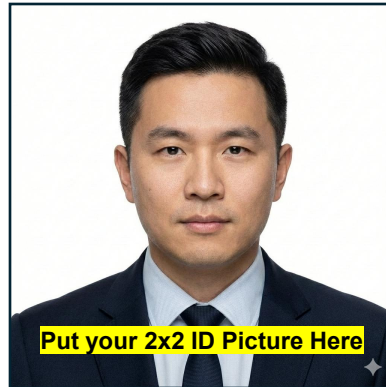
Date Signed: _____

Appendix 6
USER MANUAL

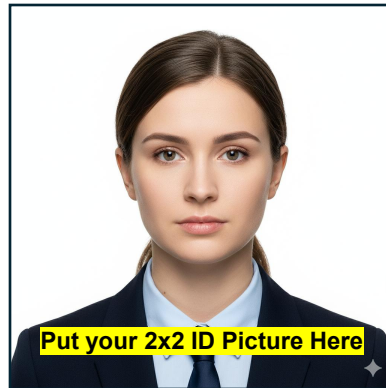
Put your content here....

Appendix 7

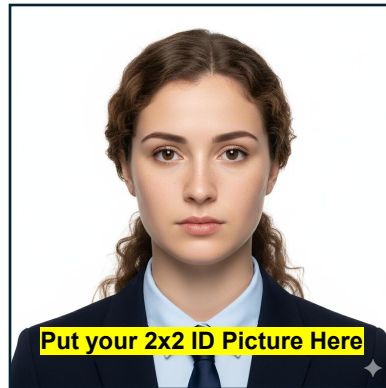
BIOGRAPHICAL STATEMENT



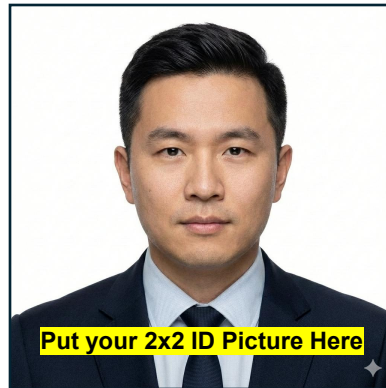
Juan Miguel Dela Cruz is a graduating student of **Bachelor of Science in Psychology** at the **Polytechnic University of the Philippines**. He completed his undergraduate coursework with a focus on developmental and educational psychology. His academic interests include student mental health, learning motivation, and psychological well-being among Filipino youth. His undergraduate research examines factors affecting academic stress among senior high school students. He plans to pursue graduate studies in psychology and engage in research related to mental health advocacy and education. (Note: Minimum of 100 and maximum of 150 words only)



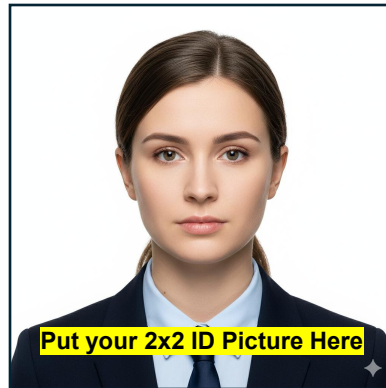
Maria Angela Dela Cruz is a graduating student of the **Bachelor of Science in Psychology** at the **Polytechnic University of the Philippines**. She completed her undergraduate studies with an emphasis on developmental and social psychology. Her academic interests include mental health awareness, emotional resilience, and psychological well-being among adolescents. Her undergraduate research focuses on stress management and coping mechanisms of senior high school students. She plans to pursue graduate studies in psychology and contribute to community-based mental health programs. (Note: Minimum of 100 words and Maximum of 150 words only)



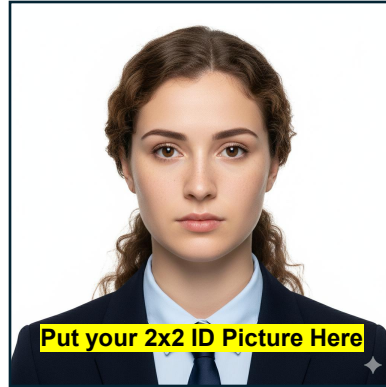
Angela Marie Santos is a graduating student of the **Bachelor of Science in Psychology** at the **Polytechnic University of the Philippines**. Her academic training centers on counseling psychology and human behavior, with particular interest in emotional intelligence and student well-being. Her undergraduate thesis explores the relationship between self-esteem and academic performance among college students. She intends to pursue professional training and graduate studies in psychology, with the goal of contributing to mental health education and support programs. (Note: Minimum of 100 and maximum of 150 words only)



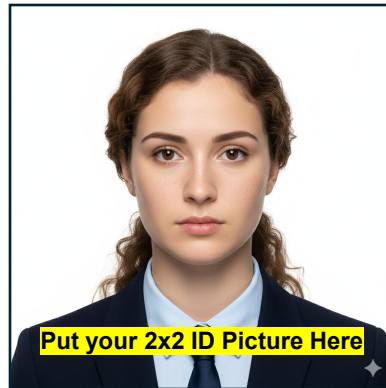
Mark Anthony Reyes is a graduating student of the **Bachelor of Science in Psychology** at the **Polytechnic University of the Philippines**. His undergraduate training emphasizes cognitive and social psychology, with particular interest in decision-making and behavioral patterns among young adults. His undergraduate thesis examines the influence of social media use on attention and academic engagement of college students. He plans to pursue graduate studies in psychology and engage in research and advocacy focused on youth mental health and well-being. (Note: Minimum of 100 and maximum of 150 words only)



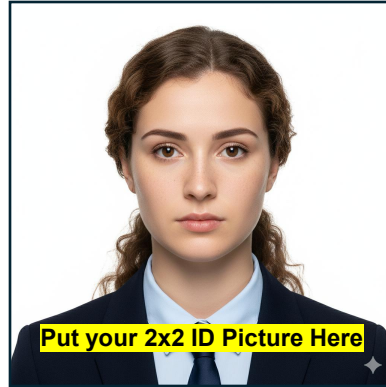
Maria Angela Dela Cruz is a graduating student of the **Bachelor of Science in Psychology** at the **Polytechnic University of the Philippines**. She completed her undergraduate studies with an emphasis on developmental and social psychology. Her academic interests include mental health awareness, emotional resilience, and psychological well-being among adolescents. Her undergraduate research focuses on stress management and coping mechanisms of senior high school students. She plans to pursue graduate studies in psychology and contribute to community-based mental health programs. (Note: Minimum of 100 and maximum of 150 words only)



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